available stacks. It is not that these existing solutions lack technical merit, but our concern lies in control. Choosing an externally developed stack means relinquishing control, hindering customisation for our customers. Having control over the stack allows one to accommodate specific requirements.

What are the possible challenges that camera manufacturers encounter?

Camera manufacturing poses its own set of challenges. While the printed circuit board assembly (PCBA) can be outsourced, the final assembly, specifically aligning the lens with the sensor, is a critical step. Even a minor deviation in alignment can result in variations in the captured image. Achieving precise alignment becomes even more crucial when producing large quantities (e.g., 10,000 pieces) that must meet strict quality standards. To address this challenge, we are investing in tools, developing our automation, and refining our manufacturing processes.

How do you see the embedded vision segment playing out in the next five years, especially in India?

The embedded vision segment will witness explosive growth with edge processing! This realisation has sparked an influx of diverse use cases, with the availability of high-quality cameras and advanced processing capabilities, over the past five years. This has led to the proliferation of algorithms, giving rise to an abundance of use cases. The government's support, coupled with the increasing prevalence of cameras, has created a fertile ground for processing and utilising visual data in areas such as traffic monitors. Vision technology is anticipated to play a crucial role in the automation of various sectors.

Electric vehicles and modern vehicles are expected to feature more cameras than their predecessors. The importance of cameras is set to surge in areas characterised by mass volume.

What role does AI play in embedded vision?

Computer vision, akin to a child, has matured into AI, a grown-up entity equipped with computer vision techniques and the capability to apply models. Initially, customers seldom ventured into processing; however, this evolved gradually. Early processing tasks included basic activities like identifying barcodes. Subsequently, computer vision emerged, incorporating features such as face recognition. Now, the current phase revolves around AI. The emphasis has shifted towards data collection and training, attracting specialised companies in these areas. Simultaneously, processors have seen continual improvement, allowing for more extensive data processing.

AI use cases can be categorised into two groups: those requiring accuracy and those that prioritise speed over accuracy. For example, providing a Facebook prompt for a new model bike recommendation allows room for error, unlike medical applications where accuracy is paramount. Although AI has progress to make in achieving accuracy, the pace of development is quicker than in earlier times.

What are the applications for this embedded vision-AI confluence?

From stadiums to retail, warehouses, and agriculture, AI is proving to be a valuable tool in streamlining processes and increasing efficiency in various industries. For instance, in the US, stadiums have camera-enabled automation in food stalls to automatically process purchases, streamlining the ordering and payment process. Similar applications are gaining traction in the retail sector. With the increasing popularity of online shopping, autonomous mobile robots (AMRs) are employed to automate tasks in warehouses, contributing to the efficiency of order fulfilment processes. AI coupled with computer vision in agriculture is used for precision farming. Tractors equipped with multiple cameras are deployed to identify specific areas, such as weeds, and apply targeted treatment. The adoption of AI in agriculture has been quicker compared to other sectors, given the controlled environment and specific applications.

While the accuracy levels in these applications may not be as critical as in certain high-precision scenarios, the speed of adoption is something to watch out for. Use cases like these demonstrate how AI complements computer vision, offering better functionalities for specific applications.

How is e-con Systems navigating through the AI revolution in embedded vision?

In the next three to four years, the anticipation is a surge in new-age AI applications. At e-con, our focus encompasses two fronts. Firstly, we aim to embed processing capabilities directly into camera hardware, envisioning compact AI-enabled cameras for various applications. We are exploring the incorporation of limited AI, such as person detection, within smaller form-factor cameras and more powerful AI in larger units. Simultaneously, we are investing in R&D to strengthen our hardware capabilities. Our projects include internal initiatives experimenting with AI models, developing use cases, and creating software stacks. For us,